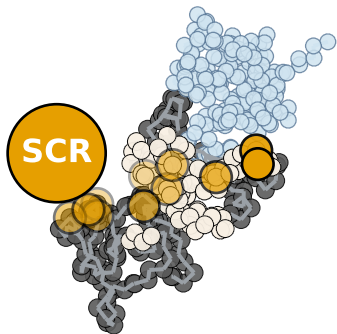
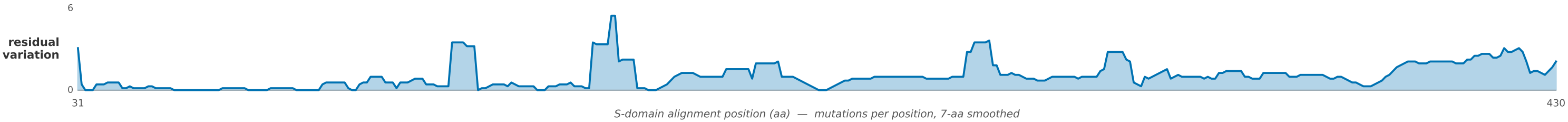


How does the relationship between HV and contact-zone variations within the β -sandwich influence switching from SI to SC?

Top-left: S-domain sequence context with residual variation. Top-right: 3D β -sandwich receptor — HV / contact-zone sites face the SCR ligand.
L1 $\rightarrow$ L2 $\rightarrow$ L3: allele definition tightens as we restrict the sites considered. L4: Objective 3 cross plan — predict SI vs SC responses from allele distance (# aa differences).



S-domain

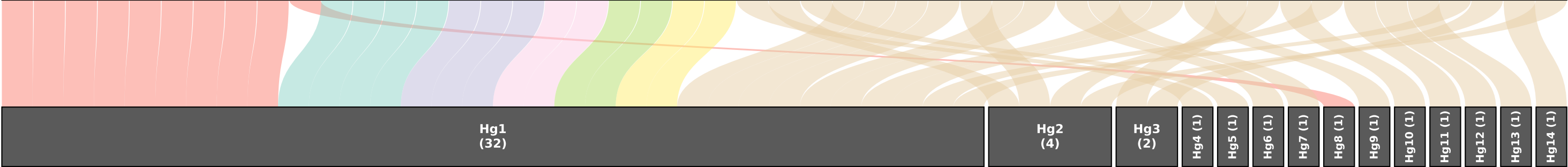


L1



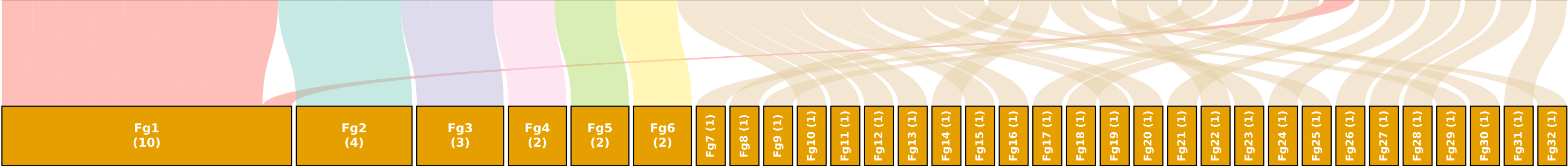
S-domain sequence identity
 $n = 49$ alleles

L2



HV-only synonymy
 $n = 14$ alleles

L3



Functional definition
 $n = 32$ alleles

cross plan
designed from L3

L4

Objective 3 · Determine allele definitions by predicting SI vs SC responses using crosses

Each arc = one candidate cross between two Fg alleles. Aim 1 (Y1): cross across three prediction tiers using AVAILABLE homozygotes (solid arcs). Aim 2 (Y2-3): generate missing homozygotes to complete the panel (dashed arcs).
Tier colour encodes the # amino-acid differences at the 223 recognition-face sites — every L3 allele appears in at least one candidate cross.

